

Watching Without Seeing Time: A Per-Question-Type Ablation of Modality Usage in a Video–Language Model

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June 2026

Abstract

Multiple-choice VideoQA accuracy is routinely read as evidence that a model *understands video*. We measure what a modern open video–language model actually uses when it answers. Using a small model-agnostic ablation harness (`vmd`), we evaluate Qwen2.5-VL-3B on a balanced 240-question subset of NExT-QA under controlled evidence conditions: full frames, no frames (blind), temporally shuffled frames (order destroyed, content preserved), and dropped frames (content destroyed). Three results follow. (i) The language prior is large: blind accuracy is 0.433 against a 0.20 chance level. (ii) The value of the video is concentrated in *descriptive* questions (+0.344 over blind) and *causal* questions (+0.200); on *temporal* questions — the category that nominally requires watching the video — frames add only +0.033. (iii) Temporal order is never used: fully shuffling the frame sequence does not reduce accuracy in any question group, while dropping 75% of the frames costs at most 0.04. On this benchmark slice the model behaves as a *bag-of-frames* system, consistent with single-frame biases reported for earlier video–language architectures. The harness, per-item predictions and a one-click reproduction notebook are public.

1 Introduction

Benchmark accuracy is an aggregate that can hide how a prediction is produced. In multimodal question answering, a model can score well by exploiting regularities of the question–answer text alone, without using the media [3]. For video, the failure has a second, subtler level: a model may genuinely use the *content* of the frames while remaining indifferent to their *order* — behaving as an image model applied to a bag of frames [2, 6].

This note asks a narrow, answerable question: *when Qwen2.5-VL-3B answers NExT-QA questions, which part of the video signal is it actually using?* We decompose the question along two axes that are usually conflated:

1. **Presence:** does accuracy drop when the frames are removed entirely (the *collapse gap* between full and blind accuracy)?
2. **Structure:** does accuracy drop when the frames are kept but their temporal order is destroyed (shuffle), versus when their content is destroyed (drop)?

and we resolve both per NExT-QA question type (causal, temporal, descriptive), since the categories make different demands on temporal understanding by construction [7].

Table 1: Accuracy of Qwen2.5-VL-3B on the balanced NExT-QA subset ($n=240$, 8 frames) under controlled evidence conditions.

Condition	Accuracy
Full (8 frames)	0.625
Blind (no frames)	0.433
Collapse gap	0.192
Shuffle 50% of frames	0.625
Shuffle 100% of frames	0.650
Drop 50% of frames	0.592
Drop 75% of frames	0.600

2 Method

Harness. `vmd` [5] treats the model as a black box behind a one-method interface: `answer(item, view)`, where the *view* controls which evidence channels the model may see and at what corruption severity. The experiment grid is therefore a property of the harness, not of the model: full evidence, blind (no media), and graded corruption of a single channel. All metrics derive from per-item correctness records, which we release alongside the code.

Frame-level conditions. Eight frames are sampled uniformly from each clip (resized to ≤ 448 px on the longest side). The two corruption operators are chosen to dissociate order from content: *shuffle*(s) permutes a fraction s of the sampled frames — the frame multiset is unchanged, only temporal structure is lost; *drop*(s) removes a fraction s of the frames — content is genuinely lost. A model that reasons over temporal structure must degrade under full shuffle; a model that only aggregates per-frame evidence will degrade under drop but not under shuffle.

Model and prompting. Qwen2.5-VL-3B-Instruct [1] with greedy decoding. The prompt presents the frames (when available), the question and the five options, and requests the option index; unparseable outputs fall back to the first option (deterministic; they are rare).

3 Experimental setup

From the NExT-QA multiple-choice test split (8,564 questions) we sample 240 questions balanced over the eight fine-grained types (30 each), grouped as causal (CW, CH; $n=60$), temporal (TN, TC, TP; $n=90$) and descriptive (DC, DL, DO; $n=90$).¹ Six conditions are evaluated: full, blind, shuffle at $s \in \{0.5, 1.0\}$ and drop at $s \in \{0.5, 0.75\}$ — 1,440 inference calls in total, no training. Chance accuracy is 0.20.

4 Results

The language prior is large. Blind accuracy is 0.433 (Table 1), more than twice the chance level. Nearly half of this benchmark slice is answerable from the question–option text alone, in

¹Annotations and videos from the `lmms-lab/NExTQA` distribution; the subset-building script (`scripts/prepare_nextqa.py`) is seeded and extracts only the required clips.

Table 2: Per-question-group breakdown. Δ columns are relative to the full condition of the same group.

Group	n	Full	Blind	GAP	Shuf. 100%	Δ shuf.	Drop 75%	Δ drop
Causal	60	0.600	0.400	+0.200	0.650	+0.050	0.567	-0.033
Temporal	90	0.511	0.478	+0.033	0.533	+0.022	0.511	0.000
Descriptive	90	0.756	0.411	+0.344	0.767	+0.011	0.711	-0.044

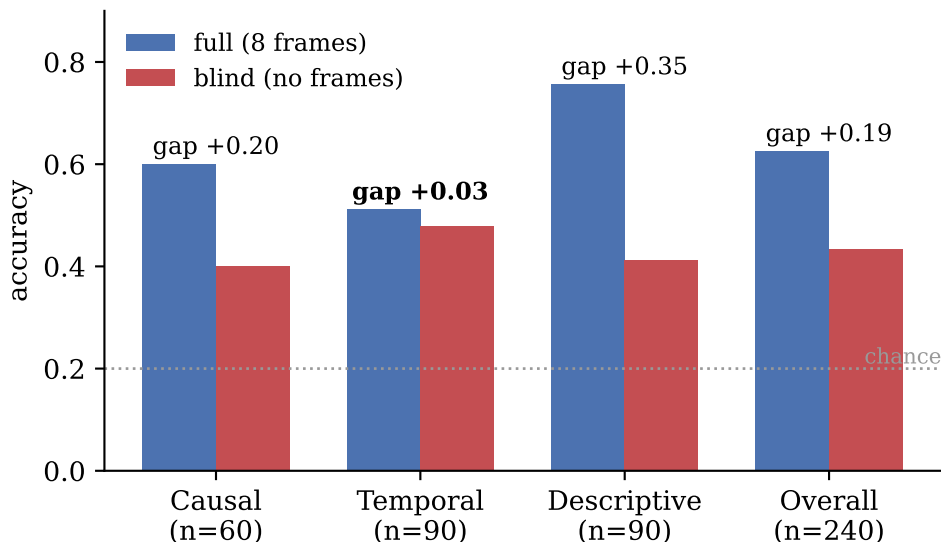


Figure 1: Full vs. blind accuracy per question group. The collapse gap — what the video actually buys — is large for descriptive questions, moderate for causal ones, and nearly zero for temporal ones.

line with prior-exploitation findings in image VQA [3]. Any headline VideoQA score on such data overstates video understanding by roughly this margin.

The video helps where it is least needed. The collapse gap is +0.344 on descriptive questions and +0.200 on causal ones, but only +0.033 on temporal questions (Table 2, Figure 1). On the category whose answers are *defined* by temporal relations — before/after, order of events — the model is barely better with the video than without it. Its temporal accuracy (0.511) is also the lowest of the three groups despite temporal blind accuracy being the *highest* (0.478): the questions are not harder as text; the model simply extracts almost nothing temporal from the frames.

Order is never used; little content suffices. Fully shuffling the frame order leaves accuracy unchanged in every group (Δ between +0.01 and +0.05, within sampling noise; Figure 2), so no measurable part of the model’s behaviour depends on temporal structure — including on temporal questions. Dropping 75% of the frames (8 $\rightarrow$ 2) costs at most 0.044: the per-frame evidence the model does use is highly redundant. Both observations match the *bag-of-frames* picture: per-frame perception aggregated order-insensitively, as reported for earlier video–language models via atemporal probes [2] and single-frame training [6], and consistent with the compositional temporal-reasoning gaps exposed by AGQA [4]. Our contribution is to isolate the same bias in a modern

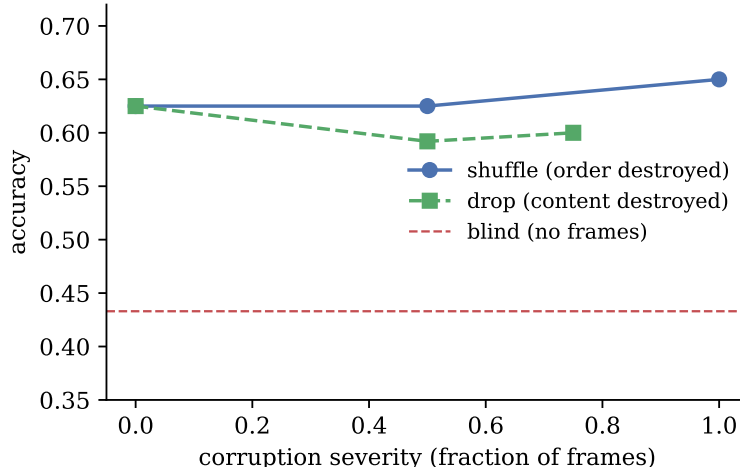


Figure 2: Accuracy under graded frame corruption (overall, $n=240$). Destroying temporal order (shuffle) has no effect; destroying content (drop) produces a modest decline. Neither approaches the blind baseline.

open VLM with a single unified ablation grid, resolved per question type, at the cost of $\sim 1,400$ inference calls and no training.

5 Limitations

$n=240$ gives roughly ± 0.06 binomial standard error per group, so small Δ s are not individually significant — the conclusions rest on the *pattern* (a near-zero temporal gap that is consistent across shuffle and drop probes), not on any single cell. Results are for one model (3B), one frame budget (8), one decoding scheme, and one benchmark; the natural next experiment is the same grid at 7B/72B to test whether scale buys temporal usage, and at 16–32 frames to rule out under-sampling as the cause of the temporal gap. Multiple-choice answer parsing, while deterministic, can interact with option phrasing. NExT-QA inherits its videos from VidOR; findings may not transfer to longer or denser footage.

6 Reproducibility

Code, the seeded subset builder, the executed notebook, per-item prediction records for every condition (`records_video.json`) and this report are available at <https://github.com/mlahozy21/video-modality-diagnostics>. The full experiment is inference-only and runs on a single L4 GPU in under an hour.

References

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